MANTA

The ocean makes the oxygen we breathe, and we fill it with plastic.

MANTA is a wave from that ocean, made by an algorithm so you print only what you need.

THE IDEA

At least half of the oxygen we breathe comes from the ocean, so MANTA is a story of life: a wave that rose from the sea floor and stayed, one continuous ribbon printed at home in 43 parts and glued without a single screw. Its forked feet are the fins of a manta, the ray that tells us how healthy the sea is. From the side, legs, seat and back draw an M, for Manta and for *Mare*. Its colour is the depth of the water, deep blue at the feet to foam at the crest, and every seam is left white, like spray.

Yet we fill that ocean with plastic; microplastics are now found even in human blood, and more and more of us, children too, print plastic at home. MANTA is plastic as well, so it is printed with sense: material only where the load is, almost no waste, recycled PETG recommended.

PRINTING — AT HOME, ALMOST NO WASTE

Every part fits a 220 × 220 × 250 mm home printer. The algorithm turns each part to the position needing least support and shapes its holes for it: supports are under 1 % of the filament, none trapped inside a hole. PETG, 0.2 mm layers, 35 % gyroid: ~16.5 kg, ~460 h on one printer.

ASSEMBLY — LIKE FURNITURE, NO SCREWS

The ribbon is cut crosswise, away from the knee and lumbar where bending peaks, so every joint is a full solid cross-section glued face to face. A printed conical peg centres each joint (0.12 mm play, tested on a printed sample); the two most-loaded joints are pinned; where the crest crosses the backrest it is glued on as a saddle. Parts are numbered; the board shows six steps.

STRENGTH AND STABILITY

Biomimicry sizes each section:

Wolff's law — thickness follows the bending moment: 84 mm at the lumbar, 32 at the crest.

Murray's law — the lumbar junction is the cube-root-sum of its branches.

Fiddlehead loop — the crest returns through the backrest: a closed loop.

Buttress roots — forked feet: a wide stance from little material.

Checked at 120 kg × 1.8 dynamic × 2.0 safety: weakest glued joint 2.1×; tipping 0.56 sideways, 0.55–0.77 rearward; 8/8 ergonomic dimensions. A full-size prototype is printed and standing — photos on the board.

~250 kg	**26×**	**~16.5 kg**	**~110 €**
safe static load	glued full-section joint	finished weight	filament + epoxy

OPEN DESIGN — THE ALGORITHM

The algorithm was written for MANTA. Choose who the chair is for and how thick it is: it rebuilds every cut, peg and joint, re-checks strength and stability, and refuses to export a chair that would fail. So anyone prints only the plastic they need: the 70 kg version built here uses half the filament. The source is open to download, change and print; manta_verify.py re-checks it all.